```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
from google.colab.patches import cv2_imshow
```

```
uploaded = files.upload()
```

Choose Files bike.jpg

bike.jpg(image/jpeg) - 93203 bytes, last modified: 9/10/2026 - 100% done

```
Saving bike.jpg to bike.jpg
```

```
filename = list(uploaded.keys())[0]

print("Uploaded file:", filename)
```

```
Uploaded file: bike.jpg
```

```
image = cv2.imread(filename)

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print("Image Shape:", gray.shape)
```

```
Image Shape: (1000, 1000)
```

```
plt.figure(figsize=(8,6))

plt.imshow(gray, cmap='gray')
plt.title("Original Grayscale Image")
plt.axis('off')

plt.show()
```




```
dft = np.fft.fft2(gray)
```

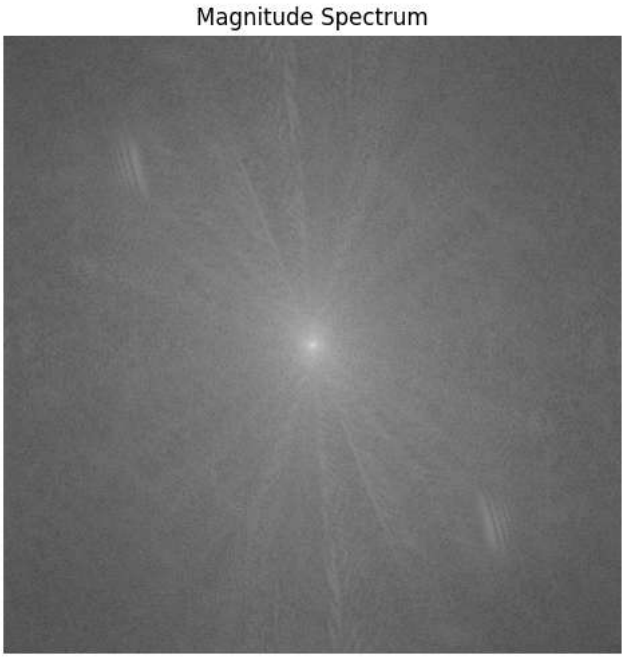
```
dft_shifted = np.fft.fftshift(dft)
```

```
magnitude_spectrum = 20 * np.log(np.abs(dft_shifted) + 1)
```

```
plt.figure(figsize=(8,6))

plt.imshow(magnitude_spectrum, cmap='gray')
plt.title("Magnitude Spectrum")
plt.axis('off')

plt.show()
```




```
rows, cols = gray.shape

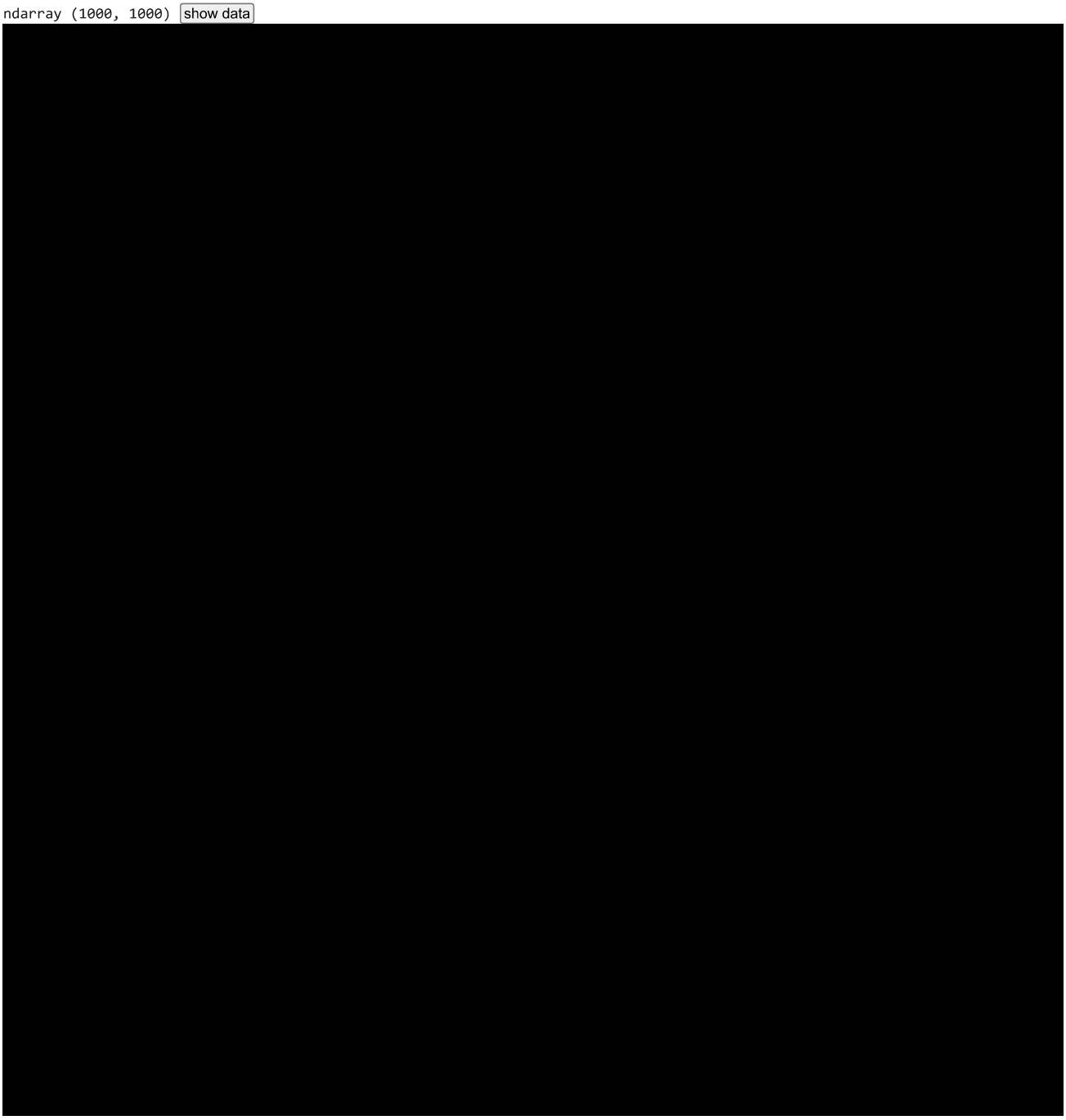
crow = rows // 2
ccol = cols // 2
```

```
low_pass_mask = np.zeros((rows, cols), np.uint8)

radius = 50

cv2.circle(
    low_pass_mask,
    (ccol, crow),
    radius,
    1,
    -1
)
```

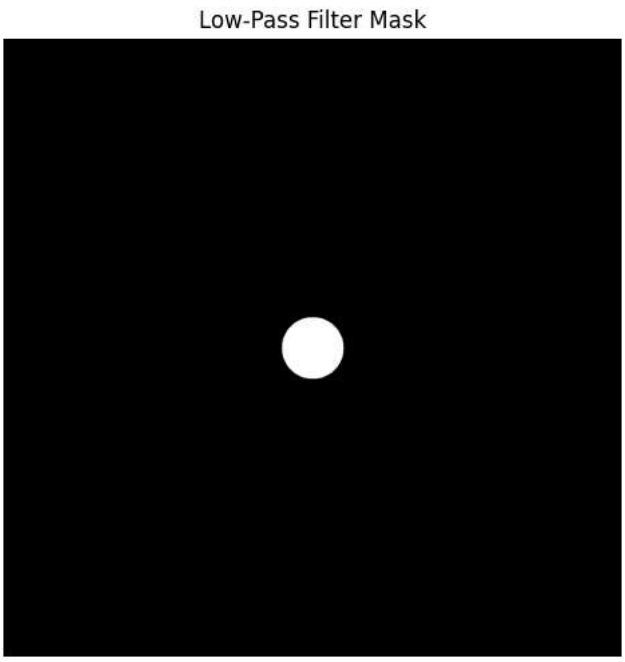
ndarray (1000, 1000) show data



```
plt.figure(figsize=(8,6))

plt.imshow(low_pass_mask, cmap='gray')
plt.title("Low-Pass Filter Mask")
plt.axis('off')

plt.show()
```




```
low_pass_dft = dft_shifted * low_pass_mask
```

```
low_pass_inverse_shift = np.fft.ifftshift(low_pass_dft)
```

```
low_pass_image = np.fft.ifft2(low_pass_inverse_shift)

low_pass_image = np.abs(low_pass_image)
```

```
low_pass_image = cv2.normalize(
    low_pass_image,
    None,
    0,
    255,
    cv2.NORM_MINMAX
).astype(np.uint8)
```

```
plt.figure(figsize=(8,6))

plt.imshow(low_pass_image, cmap='gray')
plt.title("Low-Pass Filtered Image")
plt.axis('off')

plt.show()
```




```
high_pass_mask = np.ones((rows, cols), np.uint8)

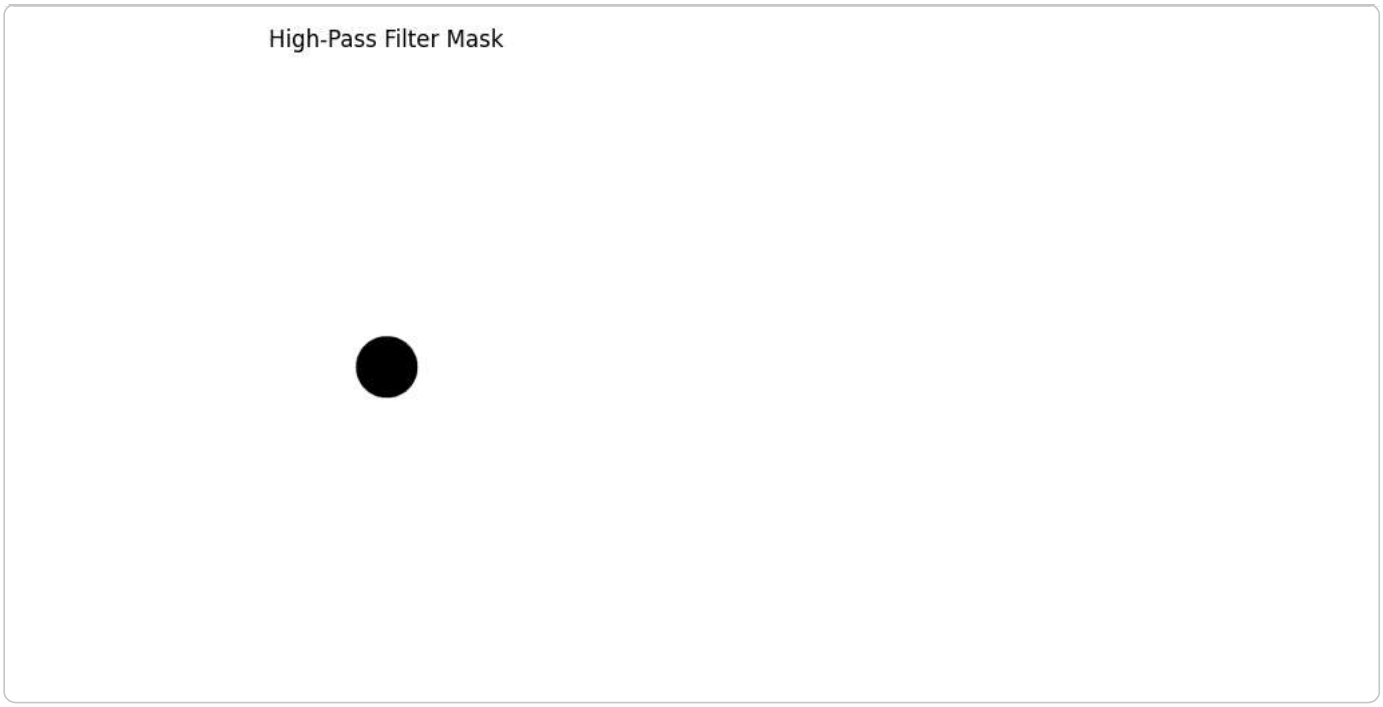
cv2.circle(
    high_pass_mask,
    (ccol, crow),
    radius,
    0,
    -1
)
```

```
ndarray (1000, 1000) hide data
array([[1, 1, 1, ..., 1, 1, 1],
       [1, 1, 1, ..., 1, 1, 1],
       [1, 1, 1, ..., 1, 1, 1],
       ...,
       [1, 1, 1, ..., 1, 1, 1],
       [1, 1, 1, ..., 1, 1, 1],
       [1, 1, 1, ..., 1, 1, 1]], dtype=uint8)
```

```
plt.figure(figsize=(8,6))

plt.imshow(high_pass_mask, cmap='gray')
plt.title("High-Pass Filter Mask")
plt.axis('off')

plt.show()
```




```
high_pass_dft = dft_shifted * high_pass_mask
```

```
high_pass_inverse_shift = np.fft.ifftshift(high_pass_dft)
```

```
high_pass_image = np.fft.ifft2(high_pass_inverse_shift)

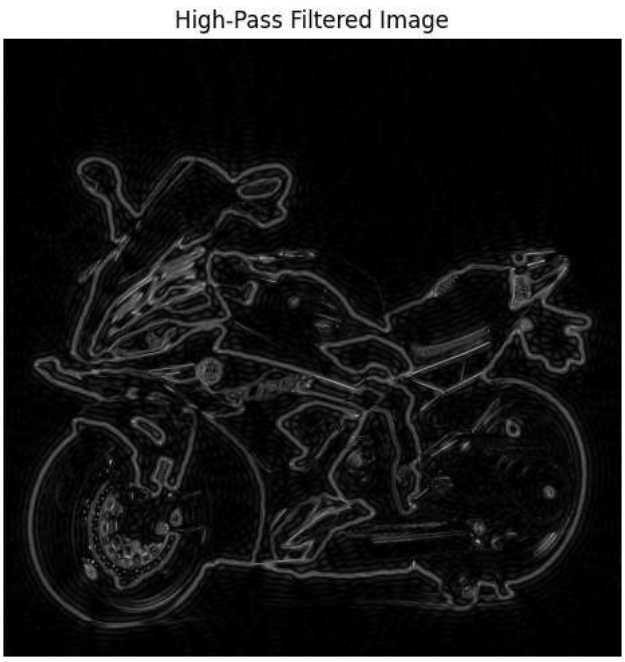
high_pass_image = np.abs(high_pass_image)
```

```
high_pass_image = cv2.normalize(
    high_pass_image,
    None,
    0,
    255,
    cv2.NORM_MINMAX
).astype(np.uint8)
```

```
plt.figure(figsize=(8,6))

plt.imshow(high_pass_image, cmap='gray')
plt.title("High-Pass Filtered Image")
plt.axis('off')

plt.show()
```




```
plt.figure(figsize=(15,10))

plt.subplot(2,3,1)
plt.imshow(gray, cmap='gray')
plt.title("Original Image")
plt.axis('off')

plt.subplot(2,3,2)
plt.imshow(magnitude_spectrum, cmap='gray')
plt.title("Magnitude Spectrum")
plt.axis('off')

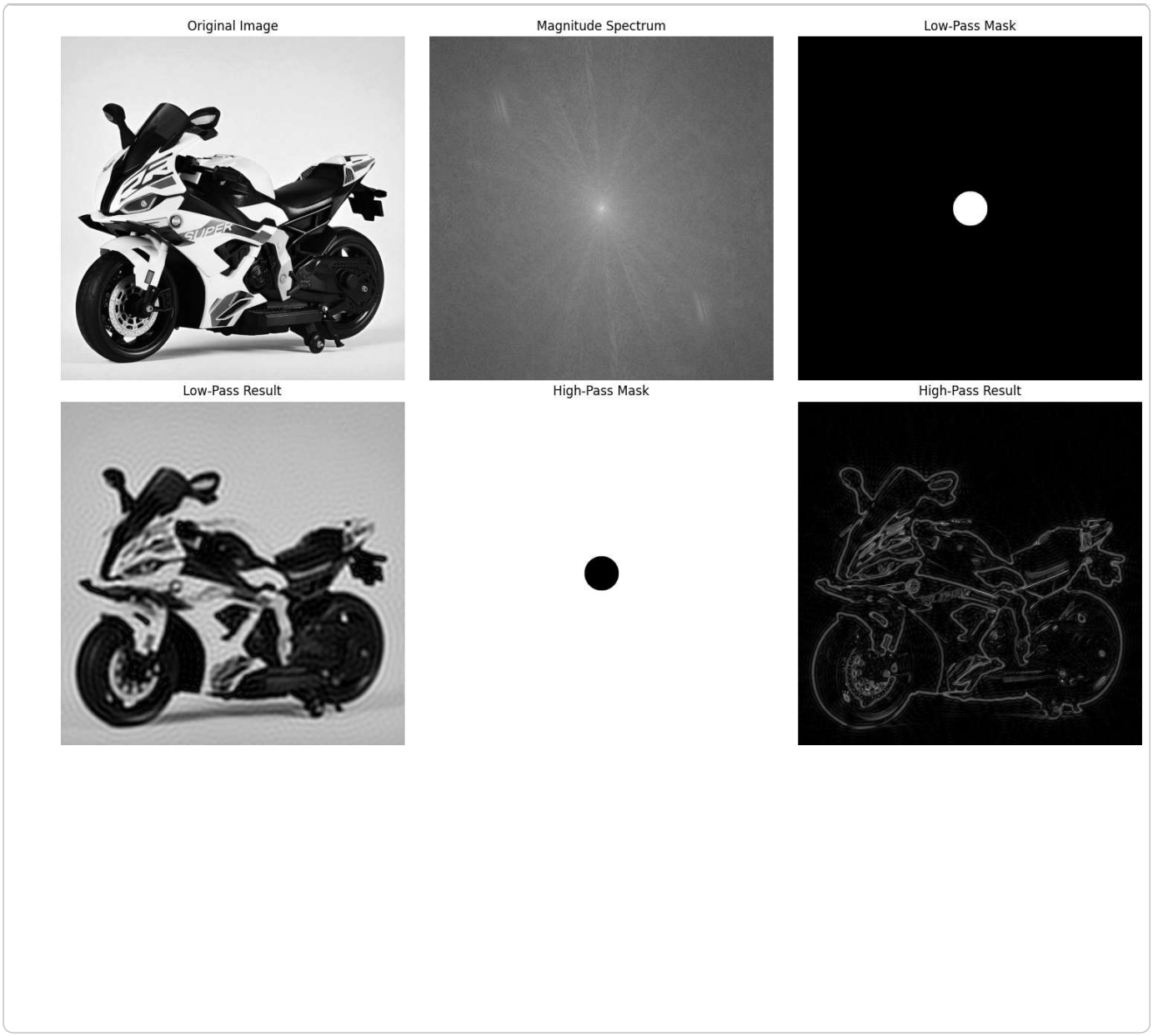
plt.subplot(2,3,3)
plt.imshow(low_pass_mask, cmap='gray')
plt.title("Low-Pass Mask")
plt.axis('off')

plt.subplot(2,3,4)
plt.imshow(low_pass_image, cmap='gray')
plt.title("Low-Pass Result")
plt.axis('off')

plt.subplot(2,3,5)
plt.imshow(high_pass_mask, cmap='gray')
plt.title("High-Pass Mask")
plt.axis('off')

plt.subplot(2,3,6)
plt.imshow(high_pass_image, cmap='gray')
plt.title("High-Pass Result")
plt.axis('off')

plt.tight_layout()
plt.show()
```



PART B — THEORY QUESTIONS

Q1. What is the Fourier Transform? Why is it important in digital image processing?

Answer:

The Fourier Transform is a mathematical technique that converts an image from the spatial domain into the frequency domain.

It represents an image in terms of its frequency components.

It is important because it allows us to:

Analyze image frequencies. Remove unwanted frequency components. Reduce noise. Enhance edges and details. Perform frequency-based filtering.

The experiment uses Fourier Transform to selectively manipulate low- and high-frequency information.

Q2. Differentiate between spatial domain and frequency domain.

Spatial Domain Frequency Domain Works directly on image pixels Works on frequency components Uses neighboring pixel values Uses frequency representation Filtering is generally performed using kernels Filtering is performed using frequency masks Examples: Gaussian, Median, Sobel Examples: Fourier Low-Pass and High-Pass filters Easy to understand for local operations Useful for selective frequency manipulation

Q3. What is the significance of the Discrete Fourier Transform (DFT) in image processing?

Answer:

DFT converts a digital image from the spatial domain into its frequency-domain representation.

It helps identify low- and high-frequency components and allows specific frequency components to be modified or removed.

It is therefore useful for:

Noise removal Image enhancement Image restoration Feature analysis Frequency-domain filtering

The experiment specifically requires calculating the DFT and analyzing its frequency spectrum.

Q4. What is the significance of shifting the zero-frequency component to the center?

Answer:

After Fourier transformation, the zero-frequency component is normally located at the corner of the frequency spectrum.

Using a frequency shift moves this component to the center.

This makes the spectrum easier to visualize and analyze because:

Low frequencies appear near the center. High frequencies appear farther from the center.

It also makes designing circular Low-Pass and High-Pass masks more convenient.

Q5. Compare Low-Pass and High-Pass Frequency Filters with suitable applications.

Low-Pass Filter High-Pass Filter Allows low frequencies to pass Allows high frequencies to pass Suppresses high frequencies Suppresses low frequencies Produces smoothing Enhances edges and fine details Useful for noise reduction Useful for edge detection Reduces fine details Highlights fine details Example: image smoothing Example: edge enhancement

The lab specifically associates Low-Pass filtering with noise reduction and High-Pass filtering with edge/detail enhancement.

Q6. What is the role of the Inverse Fourier Transform (IDFT) in image reconstruction?